

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$2(8x) + 4(7y) = 12$$

$$-2(8x) + 4(7y) = 12$$

The solution to the given system of equations is (x, y) . What is the value of $8x + 7y$?

Correct Answer: 3

Rationale

The correct answer is **3**. Adding the second equation to the first equation in the given system of equations yields $(2(8x) - 2(8x)) + (4(7y) + 4(7y)) = 12 + 12$, or $8(7y) = 24$. Dividing both sides of this equation by **8** yields $7y = 3$. Substituting **3** for $7y$ in the first equation, $2(8x) + 4(7y) = 12$, yields $2(8x) + 4(3) = 12$, or $2(8x) + 12 = 12$. Subtracting **12** from both sides of this equation yields $2(8x) = 0$. Dividing both sides of this equation by **2** yields $8x = 0$. Substituting **0** for $8x$ and **3** for $7y$ in the expression $8x + 7y$ yields $0 + 3$, or **3**. Therefore, the value of $8x + 7y$ is **3**.